

BodyMap

Medical Virtual Reality Education and Simulation

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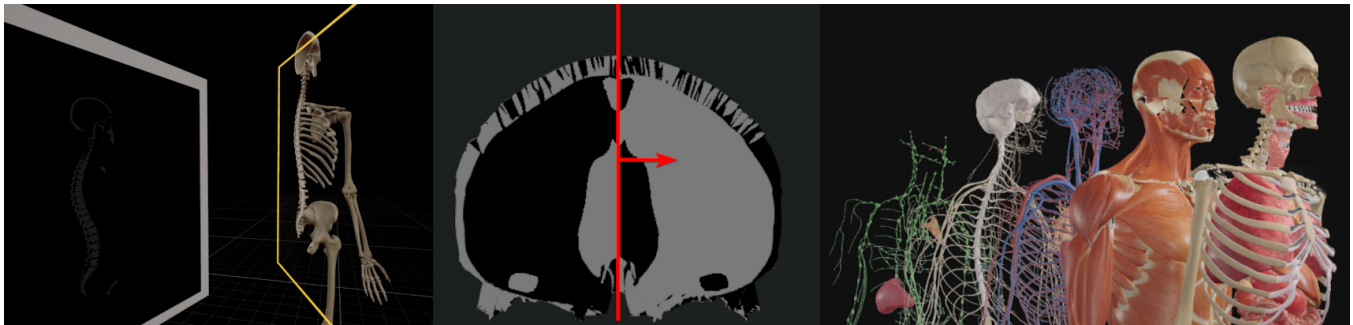


Figure 1: a) Masking tool screenshot; b) Masking tool of visible body parts; c) Body systems

ABSTRACT

BodyMap is a medically accurate representation of the human body that can be manipulated in 3D virtual reality. Users may interact with the virtual body in numerous ways, including walking into the virtual body for a detailed inspection of internal organs, grabbing out anatomy structures for a closer look, and simulating instrument insertion techniques with instant haptic feedback. This immersive learning approach enables users to transfer their knowledge gained from anatomy textbooks and cadaver dissection lessons into an immersive environment and enhance it by continuous repetition. This way of learning increases confidence in students' own abilities and knowledge.

CCS CONCEPTS

• **Human-centered computing** → **Visualization application domains.**

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SIGGRAPH '21 Immersive Pavilion, August 09-13, 2021, Virtual Event, USA

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ACM ISBN 978-1-4503-8368-4/21/08.

<https://doi.org/10.1145/3450615.3464527>

KEYWORDS

virtual virtual reality, human anatomy, medical simulation

ACM Reference Format:

Sam Jang, Tomas Bartipan, Lukas Vilim, Li-Ying Huang, Mariya Chernyavska, Ondrej Musil, Yu Chang, and Tzu-Chuen Lin. 2021. BodyMap: Medical Virtual Reality Education and Simulation. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH '21 Immersive Pavilion)*, August 09-13, 2021. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3450615.3464527>

1 INTRODUCTION

At its core every medical school aims to educate the next generation of practitioners who will make wise medical decisions and treat their patients to the best of their knowledge and abilities. As anatomy plays a critical role for all branches of medicine, it is the obligation of medical institutions and their educators to provide students with reliable and effective anatomy education. BodyMap aims to address this need with comprehensive and interactive anatomy solution for medical students. And while BodyMap does not intend to replace traditional teaching methods based on cadaver dissection, it aims to augment this educational process with advanced and engaging technology. BodyMap offers a general human anatomy learning experience that all students and residents can enjoy.

2 SYSTEM CONTENTS AND FEATURES

Our strict data reconstruction and design methods set us apart and provide users with realistic visualizations of every detail of the human body. 12 human body systems are provided in BodyMap so that users may see how the systems and

structures within those systems relate to one another. BodyMap body systems include: skin, skeleton, muscle, connective tissue, circulatory, nervous, lymphatic, digestive, reproductive, urinary, endocrine, and respiratory. BodyMap offers several interactive features to increase information retention rates through immersive visualization. Anatomy flashcards provide users with additional anatomy content, and each card is divided into different sections depending on the system to which the anatomy structure belongs. For example, when a particular muscle is selected, flashcard content is divided into sections including origin, insertion, function, innervation, and blood supply. When a nerve is selected, information on the flashcard is sectioned into origin, course, and structures innervated. Users may highlight structures within the circulatory, nervous, and lymphatic systems to observe flow pathways throughout the body.

Instruments for simulation training is also available in BodyMap. Users may simulate insertion of a medical instrument into the virtual avatar. The default instrument is configured as an 8cm blade, and can be customized into various forms of surgical tools. Haptic feedback can be felt in the controllers during virtual simulation, providing different intensities of “anti-force” during training simulations.

Magnification and visualization tools are what set BodyMap unique. Upon penetration of the virtual body during insertion simulations, the area being penetrated will be instantly magnified in real time on a separate screen, showing details including magnification of the various layers being penetrated, depth of insertion, and insertion angle.

3 TECHNOLOGY

High fidelity human body presentation in VR is a challenging task to tackle. The main issue is the sheer amount of primitives in the close vicinity to the user. Many interactions have to process a fair amount of data every frame. Our features have to be tailored to bring visually pleasing, accurate results and stand up to the performance standards of the least performant platform at the same time - for us, that is the Oculus Quest 2 standalone headset.

3.1 Body Masking

Our tools allow the users to inspect the body from inside out via masking its parts. The masking plane tool effectively cuts the body. Since regular 3D model meshes are hollow and don't look good when viewed from the inside, we need to fake this inside data to convince the user that they're indeed cutting away the body that is composed of actual, solid body parts.

Internally the tool has to compose a unique texture that covers the sliced meshes to hide the insides of the body part meshes. Which involves a separate rendering pass over the cut meshes.

The system is quite simple but has considerable performance demands which has to be compensated for via the tool's usage design, such as limiting the cut refresh frequency.

3.2 Inverse-world-movement

For performance reasons and also to simplify interaction in multiplayer, whenever the user wishes to move or scale the virtual body, we instead move or scale the user and their static surroundings with the appropriate inverse transformation.

Since our body is made up of roughly 5000 individual meshes, this provides a considerable speedup compared to moving the whole body. Since our products are targeted at standalone VR headsets (with the performance equivalent of a high-end smartphone) and our target framerate is 90fps, every millisecond counts.

As a second advantage, this approach simplifies a lot of the gameplay code, as the meshes making up the body are static and always in the same location (unless a user grabs them), simplifying transformation calculations and graphical effects.

A final advantage is multiplayer, where moving the body for all players in the classroom would be distracting and unpleasant. Users are instead free to explore the body at their own location and scale of choice and when they move the body, only their multiplayer avatar moves for the other players.